

Intraday Gas Fee Heterogeneity on Ethereum: Evidence from Operational Firms

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Abstract

Ethereum’s EIP-1559 fee mechanism was designed under the assumption of homogeneous, myopic agents responding to a single congestion signal. We examine how this assumption interacts with the heterogeneous demand structure of real-world Ethereum users. Analyzing 62,142 confirmed transactions from seven operational firms across seven industries (January–March 2026), we document significant intraday gas-fee variation: fees peak at hour 12 UTC (7 AM ET, $\hat{\beta}_{12} = \$0.054$ above the U.S. evening baseline, $p < 0.001$) and are associated with periods of elevated speculative-arbitrage activity. Operational firms exhibit heterogeneous scheduling responses moderated by transaction deferrability and gas intensity. Residual cost floors, i.e. the gap between observed expenditure and the counterfactual under perfect off-peak scheduling, range from 40.7% to 92.5% of actual expenditure, and persist even during the lowest-cost hours ($h \in \{20, 21, 22, 23\}$ UTC, 3–6 PM ET). We introduce an On-Chain Scheduling Matrix that maps firms to four scheduling regimes as a practical framework for managing gas-fee exposure under the current mechanism.

1 Introduction

Blockchain networks impose execution costs through congestion pricing. Ethereum’s EIP-1559 mechanism [BCD⁺21] raises the protocol-determined base fee whenever aggregate block demand exceeds a 50% target utilization and lowers it otherwise. [Rou21] establishes that this mechanism is incentive-compatible for myopic, value-maximizing agents: the dominant strategy is to bid true value, and the resulting allocation is ex-post efficient under independent private values.

The theoretical guarantee applies under specific scope conditions, most notably that agents are symmetric and respond myopically to a common congestion signal. Ethereum’s user population, however, includes at least two observationally distinct demand types. *Speculative-arbitrage actors* (MEV bots, DEX routers, liquidation searchers) are highly price-elastic and cluster activity during U.S. and European business hours, when arbitrage returns are largest. *Operational firms*, such as remittance platforms, healthcare coordinators, real-estate settlement systems, face contractual, counterparty, or regulatory timing constraints that limit their ability to respond to real-time price signals. These two types share network infrastructure but interact with the base-fee mechanism in qualitatively different ways.

This paper documents the cost implications of that heterogeneity. Using transaction-level on-chain data from seven firms across seven industries ($N = 62,142$ transactions, January–March 2026), we make four contributions. First, we provide the first cross-industry empirical characterization of intraday gas-fee variation from the perspective of operational users. Second, we identify the boundary conditions under which the EIP-1559 incentive-compatibility result extends or does not extend to operationally constrained participants.

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Third, we introduce a dual classification of gas fees as *execution costs in timing* (the firm chooses when to submit) and *maladaptation costs in origin* (the base fee is set by aggregate network demand outside the firm’s control), extending Transaction Cost Economics [Wil85] to the blockchain setting. Fourth, we develop an On-Chain Scheduling Matrix that translates the empirical fee schedule into actionable scheduling guidance across four firm-level regimes.

2 Boundary Conditions of the EIP-1559 Fee Mechanism

2.1 The EIP-1559 Mechanism and Its Scope

Under EIP-1559, the base fee for block b adjusts deterministically from the preceding block:

$$B_b = B_{b-1} \cdot \left(1 + 0.125 \cdot \frac{\phi_{b-1} - 0.5}{0.5} \right), \quad (1)$$

where $\phi_b = G_b^{\text{used}}/G_b^{\text{max}} \in [0, 1]$ is block fullness and the 12.5% per-block adjustment cap limits fee volatility. citeroughgarden2021 proves truth-telling is a dominant strategy for myopic agents with independent private values, yielding ex-post efficient allocation. The proof assumes agents evaluate each block independently and can freely choose whether to transact based on their valuation versus the prevailing base fee.

2.2 Boundary Conditions Not Met in the Operational Setting

Four assumptions underlying the theoretical result warrant empirical examination in the context of operational firms.

Assumption 1: Agent homogeneity. The model treats all agents as symmetric valuation-holders. In practice, speculative-arbitrage actors are highly price-elastic (entering when expected returns exceed fees) while operational firms face near-inelastic demand within their scheduling windows. This bimodal structure is not accommodated by the symmetric model.

Assumption 2: Myopic, block-by-block optimization. The incentive-compatibility proof applies to agents choosing bids one block at a time. Operational firms typically schedule transactions days or weeks in advance under contractual or regulatory constraints. Their decision is not a fee bid but a timing commitment, placing them outside the model’s strategic framework.

Assumption 3: Efficient clearing under the 50% target. Because speculative demand is bursty and concentrated in business hours globally, aggregate demand systematically exceeds the 50% target during those hours, raising base fees for all concurrent users regardless of their individual demand characteristics. Operational users who cannot defer transactions bear this congestion externality without having contributed to it.

Assumption 4: Externality resolution through pricing. The mechanism aims to clear congestion through price adjustment. However, even during the lowest-congestion hours in our sample, block-reward fullness $\hat{\phi}_{i,h^*}^{\text{br}}$ ranges from 0.184 to 0.462 across firms (Table 4). A non-zero residual congestion level persists throughout the day, representing an unresolved cost burden for operational users that scheduling discipline alone cannot eliminate.

2.3 Extended TCE Classification

Following [Wil85], transaction costs decompose into ex-ante, execution, and ex-post phases. Standard TCE treats execution costs as stable conditional on governance structure; this assumption does not hold on public blockchains, where the total transaction fee

$$C_{it} = g_{it} \cdot [B_t + \min(\pi_{it}, M_{it} - B_t)] \quad (2)$$

contains a congestion component $g_{it} \cdot B_t$ determined by aggregate network demand at time t , outside the firm’s control. We therefore propose a dual classification: gas fees are *execution costs in timing* (the firm can

influence cost through submission timing) and *maladaptation costs in origin* (the base fee reflects third-party demand the firm cannot renegotiate or contractually insulate itself from). This distinction extends the original TCE framework and is absent from prior blockchain-in-operations studies [BH20, LWS21].

3 Framework and Estimation

3.1 Intraday Cost Model

The primary estimating equation regresses realized gas fees on hour-of-day indicators in a firm-and-week fixed-effects panel:

$$\text{Gas}_{it} = \alpha + \sum_{h=0}^{22} \beta_h \mathbf{1}[H_{it} = h] + \hat{\delta} \hat{\phi}_b^{\text{br}} + \mu_i + \lambda_w + \varepsilon_{it}, \quad (3)$$

where Gas_{it} is the realized gas cost in USD; $\mathbf{1}[H_{it} = h]$ is an indicator for UTC submission hour h , with $h = 23$ as the omitted baseline; $\hat{\phi}_b^{\text{br}}$ is a block-reward fullness proxy (defined below); μ_i are firm fixed effects; λ_w are week fixed effects; and standard errors are HC3-robust throughout. The coefficients $\hat{\beta}_h$ identify the average cost premium at hour h relative to the $h = 23$ reference.

3.2 Block-Reward Fullness Proxy

Block-level congestion is measured by the clipped relative validator reward:

$$\hat{\phi}_b^{\text{br}} = \text{clip}\left(\frac{R_b}{\hat{R}^{(95)}}, 0, 1\right), \quad (4)$$

where R_b is the total validator reward for block b and $\hat{R}^{(95)}$ is the 95th percentile of block rewards pooled across all firms. The ceiling prevents extreme MEV-extraction events from compressing the scale for typical operational blocks. $\hat{\phi}_b^{\text{br}}$ enters Equation (3) as a covariate to absorb hour-level congestion variation; it is not used to estimate a structural decomposition of demand types, and all coefficients in Table 2 should be interpreted as covariate-adjusted hourly patterns rather than causal estimates.

3.3 Off-Peak Window Definition

Off-peak hours are defined throughout as $h \in \{20, 21, 22, 23\}$ UTC (3–6 PM ET). This is the unique contiguous four-hour window in which all hour-of-day coefficients from the pooled regression (Table 2, Model 1) are statistically non-positive relative to the $h = 23$ baseline: $\hat{\beta}_{20} = -0.035$ ($p < 0.001$), $\hat{\beta}_{21} = -0.021$ ($p < 0.01$), $\hat{\beta}_{22} = -0.016$ ($p < 0.01$), $\hat{\beta}_{23} = 0$ (baseline). Hours commonly labeled “off-peak” in the prior literature—such as the Asian overnight session ($h = 0$ –9 UTC)—contain significantly positive coefficients and are therefore not uniformly low-cost; the U.S. evening window is the empirically grounded choice.

3.4 Peak Shaving Score

To assess whether a firm concentrates transactions in the off-peak window beyond what uniform scheduling would predict, we compute:

$$\text{PSS}_i = \hat{s}_i^{\text{eve}} - \frac{4}{24}, \quad \hat{s}_i^{\text{eve}} = \frac{n_i^{\text{eve}}}{N_i}, \quad (5)$$

$$n_i^{\text{eve}} = \#\{t \in T_i : H_{it} \in \{20, 21, 22, 23\}\}$$

$\text{PSS}_i > 0$ indicates that firm i places more than $4/24 \approx 16.7\%$ of its transactions in the U.S. evening window, consistent with deliberate off-peak scheduling. Statistical significance is assessed via a permutation test (10,000 replications, uniform null distribution over the 24-hour clock).

3.5 Residual Cost Floor

The residual cost floor measures the gap between observed expenditure and the counterfactual under perfect off-peak scheduling:

$$R_i = C_i^{\text{actual}} - N_i \cdot \overline{\text{Gas}}_{i,h_i^*}, \quad h_i^* = \arg \min_h \overline{\text{Gas}}_{i,h}. \quad (6)$$

$R_i \geq 0$ by construction. A large ratio R_i/C_i^{actual} indicates that a firm’s cost exposure is largely structural—arising from network congestion that persists even at the cheapest available scheduling hour—rather than from suboptimal timing choices.

4 Data

Transaction records were extracted from Etherscan.io via API for January 1 through March 31, 2026. Each record includes: transaction hash; block number; Unix timestamp (UTC); originating and destination addresses; smart-contract address; transaction fee in ETH and USD; the prevailing USD-to-ETH exchange rate; and an execution-status flag. Block-level total validator reward R_b was downloaded separately for each block appearing in the transaction data.

Terminology. Ethereum has operated under proof-of-stake since September 2022 (“the Merge”). Blocks are produced by *validators*; all prior references to “miner” or “block miner” have been updated accordingly.

UTC-to-ET conversion. U.S. clocks advanced to Eastern Daylight Time (UTC−4) on March 8, 2026. For January 1–March 7, hour 15 UTC corresponds to 10 AM EST; from March 8 onward, to 11 AM EDT. All Eastern Time labels in this paper reflect the correct daylight-saving adjustment.

The final sample contains $N = 62,142$ confirmed transactions after removing failed transactions (`isError`= 1) and records with missing required fields (Table 1).

Table 1: Sample Firms and Transaction Counts

Industry	Firm	N_i
Technology	Coins.ph	54,651
Finance	Anchorage Digital	1,785
Healthcare	Solve.Care	116
Supply Chain	Morpheus.Network	756
Real Estate	Propy Inc.	4,290
Consumer Goods	Nike (Ondo)	72
Admin Services	BrainTrust (BTRST)	472
Total		62,142

Records extracted from Etherscan.io, January 1–March 31, 2026. N_i : confirmed transaction count after data cleaning. Coins.ph contributes 88% of observations; robustness checks using equal firm weighting are reported in the supplementary material.

5 Results

5.1 Pooled Intraday Cost Structure

Table 2 reports the pooled hour-of-day regression.

Peak hour. The highest coefficient in Model (1) is $\hat{\beta}_{12} = 0.054$ ($t = 13.59$, $p < 0.001$), placing the cost peak at hour 12 UTC (7 AM ET), followed by $\hat{\beta}_{13} = 0.045$ ($t = 11.70$, $p < 0.001$) at hour 13 UTC (8 AM ET). This peak coincides with the overlap of European and early U.S. trading sessions.

Baseline. The intercept $\hat{\alpha} = 0.1597$ ($t = 48.64$, $p < 0.001$) is the expected gas cost at the $h = 23$ baseline (6 PM ET, U.S. evening).

Explanatory power. Adjusted R^2 is 1.1% in Model (1) and 2.5% in Model (2), reflecting substantial within-hour fee volatility driven by block-level congestion shocks not captured by hour indicators alone.

Fullness pass-through. The block-reward fullness coefficient $\hat{\delta} = 0.123$ ($t = 12.45$, $p < 0.001$) in Model (2) indicates that higher block-reward fullness is associated with higher gas costs conditional on hour-of-day. This association is consistent with speculative-arbitrage activity elevating congestion during business hours, though the observational design does not support a causal decomposition of demand types. Establishing such a decomposition—for example, via MEV-Boost block identification or exogenous NFT-drop demand shocks as natural experiments—is a direction for future work.

Table 2: Hour-of-Day Gas Fee Regressions, Pooled Sample

Hour h (UTC) [ET]	Model (1): Base		Model (2): $+\hat{\phi}_b^{\text{br}}$	
	$\hat{\beta}_h$	t	$\hat{\beta}_h$	t
$h = 0$ [7 PM]	0.0087	1.681	0.0072	1.383
$h = 1$ [8 PM]	0.0340	4.277***	0.0341	4.315***
$h = 2$ [9 PM]	0.0095	1.273	0.0162	2.099*
$h = 3$ [10 PM]	0.0103	2.626**	0.0164	4.042***
$h = 4$ [11 PM]	0.0330	6.725***	0.0400	7.946***
$h = 5$ [12 AM]	0.0150	3.855***	0.0231	5.680***
$h = 6$ [1 AM]	0.0255	6.501***	0.0313	7.703***
$h = 7$ [2 AM]	0.0289	7.406***	0.0336	8.340***
$h = 8$ [3 AM]	0.0272	6.923***	0.0277	6.865***
$h = 9$ [4 AM]	0.0278	7.192***	0.0284	7.134***
$h = 10$ [5 AM]	0.0401	10.433***	0.0410	10.334***
$h = 11$ [6 AM]	0.0396	10.035***	0.0372	9.121***
$h = 12$ [7 AM]	0.0540	13.589***	0.0501	12.257***
$h = 13$ [8 AM]	0.0448	11.701***	0.0361	8.996***
$h = 14$ [9 AM]	0.0386	5.959***	0.0226	3.535***
$h = 15$ [10 AM]	0.0436	6.461***	0.0240	3.583***
$h = 16$ [11 AM]	0.0391	4.707***	0.0199	2.486*
$h = 17$ [12 PM]	-0.0492	-10.320***	-0.0623	-12.676***
$h = 18$ [1 PM]	-0.0033	-0.371	-0.0152	-1.735
$h = 19$ [2 PM]	-0.0014	-0.099	-0.0095	-0.711
$h = 20$ [3 PM]	-0.0353	-4.074***	-0.0433	-5.079***
$h = 21$ [4 PM]	-0.0206	-2.682**	-0.0221	-2.926**
$h = 22$ [5 PM]	-0.0158	-2.901**	-0.0171	-3.094**
$\hat{\delta}(\hat{\phi}_b^{\text{br}})$	—	—	0.1230	12.445***
$\hat{\alpha}$ ($h = 23$, 6 PM ET)	0.1597	48.636***	0.1267	29.508***
Adj. R^2	0.0107		0.0254	
N	62,142		62,142	

Dependent variable: gas fee in USD per transaction. $h = 23$ UTC (6 PM ET) is the omitted baseline. Shaded rows: cost peak at $h = 12$ (bold) and U.S. evening off-peak window $h \in \{20, 21, 22, 23\}$ (statistically non-positive coefficients). ET labels reflect EST (UTC-5) for January 1–March 7 and EDT (UTC-4) from March 8, 2026. HC3-robust standard errors. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

5.2 Firm-Level Scheduling Behavior and Residual Floors

Table 3 reports Peak Shaving Scores, fee savings, and residual cost floors for all seven firms. Table 4 reports the firm-level residual floor decomposition confirming that non-zero congestion persists at each firm’s cheapest scheduling hour.

Three findings emerge from Tables 3 and 4.

Scheduling patterns are externally constrained. Firms with positive PSS values—Solve.Care (+0.023), Coins.ph (+0.003), and Nike/Ondo (0.000, neutral)—have geographic or operational characteristics that

Table 3: Cross-Industry Summary: Scheduling Scores, Residual Floors, and Fullness Pass-Through

Industry	Firm	N_i	n_i^{eve}	$\hat{s}_i^{\text{eve}}$	PSS $_i$	Fee savings	Floor $_i^{\text{USD}}$	Floor% $_i$	$\hat{\delta}_i$
Healthcare	Solve.Care	116	22	0.190	+0.023	−26.4%	\$19.46	64.1%	0.176
Technology	Coins.ph	54,651	9,284	0.170	+0.003	5.3%	\$4,209.29	43.2%	0.079***
Consumer Goods	Nike (Ondo)	72	12	0.167	0.000	60.7%	\$5.40	92.5%	−0.245
Supply Chain	Morpheus	756	120	0.159	−0.008	35.5%	\$67.20	67.0%	0.243***
Real Estate	Propy	4,290	669	0.156	−0.011	15.2%	\$512.66	40.7%	0.578***
Admin Services	BrainTrust	472	72	0.153	−0.014	6.7%	\$21.43	47.3%	0.024
Finance	Anchorage	1,785	267	0.150	−0.017	59.7%	\$30.02	86.8%	0.069***
Pooled		62,142		0.167	—	—	—	—	0.123***

Off-peak window: $h \in \{20, 21, 22, 23\}$ UTC (3–6 PM ET, U.S. evening). n_i^{eve} : transactions in off-peak window. $\hat{s}_i^{\text{eve}} = n_i^{\text{eve}}/N_i$. PSS $_i = \hat{s}_i^{\text{eve}} - 4/24$ (benchmark $\approx 16.7\%$; positive values indicate above-benchmark concentration in the U.S. evening window). Fee savings = $(\overline{\text{Gas}}_{i,\text{peak}} - \overline{\text{Gas}}_{i,\text{eve}})/\overline{\text{Gas}}_{i,\text{peak}}$ (negative for Solve.Care reflects high-gas outliers in certain overnight hours). Floor $_i^{\text{USD}}$: residual cost under perfect off-peak scheduling (Equation 6). $\hat{\delta}_i$: fullness pass-through from firm-level regression; Nike (Ondo) negative value reflects a timing pattern associated with price-event clustering, interpreted with caution given $N = 72$. *** $p < 0.001$.

Table 4: Residual Cost Floor by Firm

Industry	Firm	h_i^*	$\overline{\text{Gas}}_{i,h^*}$	C_i^{actual}	R_i	Floor% $_i$	$\hat{\phi}_{i,h^*}^{\text{br}}$
Cons. Goods	Nike (Ondo)	6	\$0.006	\$5.84	\$5.40	92.5%	0.462
Finance	Anchorage	8	\$0.003	\$34.59	\$30.02	86.8%	0.184
Supply Chain	Morpheus	3	\$0.044	\$100.31	\$67.20	67.0%	0.206
Healthcare	Solve.Care	1	\$0.094	\$30.38	\$19.46	64.1%	0.209
Admin Svcs	BrainTrust	3	\$0.051	\$45.35	\$21.43	47.3%	0.279
Technology	Coins.ph	20	\$0.101	\$9,749.62	\$4,209.29	43.2%	0.355
Real Estate	Propy	5	\$0.174	\$1,259.63	\$512.66	40.7%	0.203

h_i^* : UTC hour with lowest historical mean gas fee for firm i . $\hat{\phi}_{i,h^*}^{\text{br}}$: mean block-reward fullness at h_i^* ; all values are strictly positive, indicating that non-zero network congestion persists even at the lowest-cost scheduling hour for each firm. Rows sorted by Floor% $_i$ descending. All monetary values in USD.

allow transaction timing toward the U.S. evening window. Firms with negative scores face external timing constraints: institutional settlement cycles (Anchorage Digital, −0.017), property closing deadlines (Propy, −0.011), governance cycles (BrainTrust, −0.014), or IoT-oracle triggers (Morpheus.Network, −0.008). The variation is consistent with operational heterogeneity moderating scheduling capacity, though the seven-firm sample precludes formal regression on these moderators.

Residual cost floors are structural. Even the best-performing scheduler in the sample (Solve.Care, PSS = +0.023) faces a residual floor equal to 64.1% of actual expenditure. Floors range from 40.7% to 92.5% across firms. As Table 4 shows, $\hat{\phi}_{i,h^*}^{\text{br}}$ is strictly positive for every firm (0.184–0.462), confirming that non-zero network congestion persists at the cheapest available scheduling hour for each firm in the sample. The residual floor therefore reflects structural network conditions rather than suboptimal firm behavior.

Pass-through heterogeneity is consistent with asset specificity. The pass-through coefficient $\hat{\delta}_i$ varies from −0.245 to 0.578 across firms, a pattern consistent with the TCE dimension of asset specificity [Wil85]: Propy’s real-estate contract interactions carry the highest pass-through; Anchorage’s standardized custody transfers carry low pass-through; BrainTrust’s administrative calls are statistically indistinguishable from zero. The observational design does not establish causation; the pattern warrants further investigation with a larger cross-firm sample.

6 The On-Chain Scheduling Matrix

Two firm-level dimensions jointly determine the scope for gas-fee reduction through transaction timing. *Deferrability* $d_{it} \in \{0, 1\}$: whether transaction t can be submitted in any block within a window $[t, t + \Delta]$ without breaching a contractual, regulatory, or operational deadline. *Gas intensity* g_{it} : the computational weight of the transaction, which scales the dollar saving from submitting in a lower-cost hour.

Table 5 maps these two dimensions to four scheduling regimes. Hour recommendations are derived directly from Table 2: off-peak hours are $h \in \{20, 21, 22, 23\}$ UTC (the four hours with statistically non-positive coefficients); the peak-hour premium for budgeting purposes is $\hat{\beta}_{12} = \$0.054$ (hour 12 UTC, empirical cost peak) and the off-peak baseline is $\hat{\alpha} = \$0.160$ ($h = 23$ intercept, Table 2).

Table 5: On-Chain Scheduling Matrix

	High gas intensity ($g_{it} > \bar{g}$)	Low gas intensity ($g_{it} \leq \bar{g}$)
High defer- rability ($d_{it} = 1$)	<i>Regime I: Full peak shaving.</i> Schedule deferrable transactions to $h \in \{20, 21, 22, 23\}$ UTC (3–6 PM ET), where $\hat{\beta}_h \leq 0$. Estimated saving: up to \$0.054/transaction relative to the hour-12 cost peak (baseline \$0.160). Batch submissions into nightly windows to amortize scheduling overhead. <i>Exemplar:</i> Morpheus.Network ($\hat{\delta} = 0.243^{***}$).	<i>Regime II: Selective peak shaving.</i> Submit during $h \in \{20, 21, 22, 23\}$ UTC when operationally feasible. Because gas intensity is low, per-transaction savings are modest; the scheduling overhead is warranted only when transaction volume is sufficient for aggregate savings to exceed implementation costs. <i>Exemplar:</i> Coins.ph ($\hat{\delta} = 0.079^{***}$).
Low defer- rability ($d_{it} = 0$)	<i>Regime III: Cost provisioning.</i> Timing optimization is infeasible given external constraints. Instead, pre-allocate gas budgets using Table 2 as a forward cost curve: budget an additional \$0.054 per high-gas transaction submitted during $h \in \{11, \dots, 16\}$ UTC (6 AM–11 AM ET) relative to the \$0.160 evening baseline. <i>Exemplar:</i> Propy Inc. ($\hat{\delta} = 0.578^{***}$).	<i>Regime IV: Accept market rate.</i> Both non-deferrable and low gas intensity; the expected cost differential between peak and off-peak windows is small in absolute terms. Operational resources are better directed toward ex-ante cost reduction (contract optimization, calldata compression) than timing controls. <i>Exemplar:</i> Anchorage Digital ($\hat{\delta} = 0.069^{***}$).

Peak-hour premium: $\hat{\beta}_{12} = \$0.054$ (Table 2, $h = 12$ UTC = 7 AM ET, $p < 0.001$). Off-peak baseline: $\hat{\alpha} = \$0.160$ ($h = 23$ intercept, Table 2). Off-peak window: $h \in \{20, 21, 22, 23\}$ UTC (3–6 PM ET)—the four consecutive hours with statistically non-positive hour-of-day coefficients in the pooled regression.

7 Conclusion

We document significant intraday gas-fee variation on Ethereum using transaction-level data from seven operational firms across seven industries (January–March 2026, $N = 62,142$). Fees are highest at hour 12 UTC (7 AM ET, $\hat{\beta}_{12} = \$0.054^{***}$ above the U.S. evening baseline of \$0.160), a pattern that co-occurs with elevated block-reward fullness and is consistent with speculative-arbitrage activity elevating congestion costs for concurrent operational users. Residual cost floors ranging from 40.7% to 92.5% of actual expenditure persist even under optimal scheduling to the U.S. evening off-peak window ($h \in \{20, 21, 22, 23\}$ UTC), arising from non-zero network congestion that is present throughout the day.

These findings are relevant to the mechanism-design literature on EIP-1559 [Rou21] in the following sense. The incentive-compatibility guarantee applies under a specific set of agent assumptions (myopia, symmetry, and independent private values) that may not hold for operationally constrained firms whose transaction timing is governed by contractual or regulatory schedules rather than by real-time fee comparisons. Understanding the cost implications of this heterogeneity, and whether mechanism modifications could reduce the residual burden on low-elasticity users, is a productive direction for future mechanism-design research.

The On-Chain Scheduling Matrix provides a practical framework for firms operating under the current mechanism: four regimes mapping transaction deferrability and gas intensity to scheduling strategies, grounded in the empirical fee schedule of Table 2.

Three directions merit priority in future work. *Causal identification*: natural experiments such as MEV-Boost block classification or exogenous NFT-drop demand shocks would allow the associational findings reported here to be strengthened into causal estimates of speculative demand’s contribution to operational user costs. *Mechanism extensions*: whether a modified base-fee rule conditioning on observable demand-type composition could better accommodate heterogeneous user populations without sacrificing the myopic incentive-compatibility guarantee. *Layer-2 counterfactual*: whether the intraday cost structure documented here attenuates on Layer-2 networks where MEV extraction is architecturally constrained, providing a within-firm test of the speculative-demand mechanism.

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